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Reforming the Electricity Grid for a Sustainable Energy Transition: Lessons from the United Kingdom

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Abstract: The electricity grid is the physical backbone of the low-carbon transition, yet in most industrialised economies it was designed for a fossil-fuelled, unidirectional, and centralised system. This paper examines how the electricity grid of Great Britain has become the object of a comprehensive reform programme and asks what generalisable lessons its experience offers for other countries pursuing deep decarbonisation. A qualitative policy-document analysis, triangulated with publicly available quantitative indicators, is adopted: policy documents, regulatory determinations, and network operator business plans issued between 2020 and 2026 are analysed and triangulated with published quantitative indicators, including connection queue volumes, transmission constraint costs, and planned capital expenditure. The evidence shows that the pre-reform "first come, first served" model produced a connection queue of over 700 GW of generation and storage and more than 120 GW of demand by 2025, while transmission constraint costs reached approximately £1.9 billion in 2024/25. In response, three mutually reinforcing reform strands have been introduced: a redesigned connections process (Target Model Option 4+, TMO4+) that combines readiness with strategic alignment; a step change in transmission investment under the Accelerated Strategic Transmission Investment (ASTI) framework and the RIIO-T3 (Revenue = Incentives + Innovation + Outputs, transmission price control 2026–2031) price control; and a strengthened regulatory regime with enforceable delivery obligations. This policy synthesis concludes that reforming the grid is as much an institutional and regulatory task as a technical one, and that coordinated action across the system operator, the regulator, government and network companies is a necessary condition for delivering an affordable, secure and low-carbon electricity system.

Keywords: Electricity grid reform; Connections queue; Transmission investment; Clean Power 2030; RIIO-T3; National Energy System Operator; Constraint costs; Energy sustainability

1 Introduction

Electricity is central to modern economies, and its share of final energy consumption is expected to rise substantially over the coming decades as heating, transport and industry electrify [1, 2]. The International Energy Agency has estimated that global grid infrastructure must grow by more than a fifth in length by 2030, and that annual grid investment must roughly double, if national and international climate commitments are to be met [3]. The grid, in other words, is no longer a passive utility asset; it is a strategic enabler of decarbonisation, energy security and economic growth [4, 5].

Great Britain (GB) provides a particularly instructive case study. Its transmission network, one of the oldest large-scale grids in the world, was designed in the mid-twentieth century for a system dominated by coal- and gas-fired stations located close to industrial and urban demand [6]. Over the past decade the location, scale and character of both generation and demand have changed rapidly. Wind now supplies roughly 30 percent of GB electricity, with most capacity concentrated in Scotland and the North Sea, while demand centres remain in England [7]. Simultaneously, connection applications from batteries, data centres, gigafactories and electrified industrial sites have surged, resulting in a connection queue that reached 722 GW of generation and storage and 125 GW of demand by mid-2025 several times the physical capacity the country is expected to need in 2030 [8, 9].

These pressures triggered a comprehensive reform programme covering the connections process, the regulatory price control framework, the strategic planning of infrastructure and the accountability of network companies [10, 11]. Recent government analysis has argued that without such reform, network delays would become the binding constraint

on clean power delivery and would materially raise system costs [12]. This paper examines those reforms, evaluates their emerging outcomes, and draws implications for other jurisdictions.

The remainder of the paper is organised as follows. Section 2 reviews the literature on grid reform for decarbonisation. Section 3 sets out the methodology. Section 4 documents the drivers of reform in GB. Section 5 analyses the reform architecture across the connections queue, transmission investment, and regulation. Section 6 discusses cross-cutting lessons and situates GB experience against European practice. Section 7 draws policy implications, and Section 8 concludes.

2 Literature Review

The literature on electricity network transformation for a low-carbon energy system spans three broad strands: technical modernisation, market and regulatory design, and infrastructure planning under uncertainty. On the technical side, a substantial body of work has explored the integration of variable renewable energy through smart grid technologies, high-voltage direct current (HVDC) transmission, energy storage and grid-forming inverters [13–16]. Recent reviews highlight the growing importance of artificial intelligence and digital twins for real-time monitoring, congestion management and asset planning [17, 18].

A parallel literature examines market and regulatory reform. Newbery [19] and as documented in the standard reference on electricity market design [20] have argued that decarbonisation exposes the limits of purely energy-only markets designed for dispatchable thermal fleets, and that regulators must combine long-term capacity signals, locational pricing and network access reform. Grubb and Newbery [12] emphasise that the interaction between network regulation, wholesale market design and support schemes must be treated holistically. In the European Union, the European Court of Auditors (ECA) has concluded that current network investment plans lag the requirements of the 2050 climate-neutrality target and estimated an aggregate need for around €2 trillion of grid investment [21, 22]. Comparable concerns have been raised in the United States about transmission planning and interconnection queue reform [23, 24].

The third strand concerns infrastructure planning under deep uncertainty. Fürsch et al. [25] and van der Weijde and Hobbs [26] developed early stochastic models of European transmission expansion. More recent work by Brown and Reichenberg [27] uses linear-programming energy system models to co-optimise generation and transmission investment across large regions, showing consistently that interconnected grids reduce total system cost.

While these strands are well developed, comparatively little peer-reviewed work has synthesised the recent GB reform package, most of which has been implemented since 2024 through regulatory rather than legislative instruments. This paper contributes to closing that gap.

3 Methodology

This study adopts a mixed-methods approach that combines qualitative document analysis with descriptive quantitative synthesis. The evidence base comprises three categories of source. First, official policy documents were reviewed, including the Clean Power 2030 Action Plan and its Connections Reform Annex [10, 28], the joint government regulator open letter on aligning connections with strategic plans [11], and the Statutory Security of Supply Report 2024 [29]. Second, regulatory determinations and consultation outputs were analysed, notably Office of Gas and Electricity Markets (Ofgem)'s RIIO-T2 (Revenue = Incentives + Innovation + Outputs for Transmission, 2021–2026) electricity transmission annual reports and RIIO-T3 (the third transmission price control period, 2026–2031) final determination [30–32], the Accelerated Strategic Transmission Investment (ASTI) framework [33], and the end-to-end connections review [34]. Third, industry sources were used, including the National Energy System Operator (NESO) publications on connections reform and the Beyond 2030 network study [35, 36], and network operator business plans and supporting materials [37–39].

Sources were selected purposively against three criteria: (i) formal status as an official policy, regulatory or system-operator publication rather than commentary or opinion pieces; (ii) direct relevance to one or more of the three reform pillars examined in this study; and (iii) public availability, so that the analysis can be reproduced by other researchers. Where multiple versions of a document existed, for example successive updates to the Clean Power 2030 Action Plan, the most recent publicly available version at the time of writing (February 2026) was used, and superseded versions were noted where their content diverged materially. This criteria-based approach to source selection follows established conventions in document-analysis research [40].

Quantitative indicators were extracted directly from these sources and, where necessary, from published trade press analyses that cite primary regulatory data. The indicators focus on four dimensions: (i) size and composition of the connections queue; (ii) transmission constraint and balancing costs; (iii) planned capital expenditure; and (iv) delivery output metrics such as circuit kilometres and connected capacity. Where estimates differ across sources, the range is reported.

The analytical strategy was thematic. Reforms were coded against the three interdependent domains identified in the literature: (a) the connections process, (b) the transmission investment framework, and (c) the regulatory

accountability regime. Cross-cutting themes strategic coordination, cost allocation, and delivery risk were then identified inductively. This approach is transparent and reproducible but is subject to two limitations inherent in document-based research: first, reform outcomes can only be assessed on early-stage evidence; and second, the analysis relies on publicly disclosed data, which may not fully capture implementation challenges or commercially sensitive delivery risks faced by network companies.

4 Drivers of Grid Reform in Great Britain

4.1 Decarbonisation and the Clean Power 2030 Mission

The United Kingdom has legally committed to net-zero greenhouse gas emissions by 2050, and the current government has adopted a more immediate mission of achieving a clean-power electricity system by 2030 [10, 41]. Clean Power 2030 requires approximately 43–50 GW of offshore wind, 27–29 GW of solar, 27 GW of onshore wind and around 27 GW of storage to be operational by the end of the decade [10]. Delivering this pipeline entails not only building new generation but also, and more critically, ensuring that the transmission and distribution networks can transport the resulting flows.

4.2 Rapid Growth of the Connections Queue

The scale of the connection challenge is illustrated in Figure1. Between 2020 and 2025, the volume of contracted generation and storage in the GB queue increased roughly tenfold, while the demand-side queue tripled between late 2024 and mid-2025 as a wave of data centre, storage and industrial applications arrived [8]. By September 2024 the combined queue stood at 722 GW, several times the physical capacity the country is expected to require in 2030 [11].

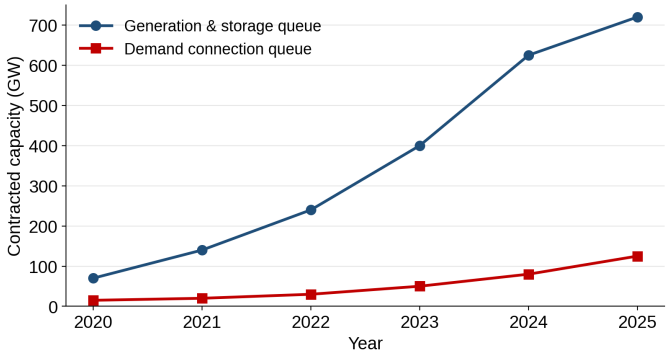


Figure 1. Growth of the Great Britain electricity connections queue, 2020–2025, adapted from [8, 11]

4.3 Rising Constraint and Curtailment Costs

Because most new renewable capacity connects in Scotland while demand is concentrated in southern England and the Midlands, insufficient boundary transfer capability across the B4 and B6 constraint boundaries has forced NESO to pay wind farms to reduce output while dispatching thermal plant elsewhere. Constraint costs have risen sharply, as shown in Figure 2. Independent trackers estimate that in 2025 approximately £380 million was paid to wind farms to reduce output and £1.08 billion to replacement gas plant, taking total wind-related curtailment costs to around £1.5 billion [42, 43].

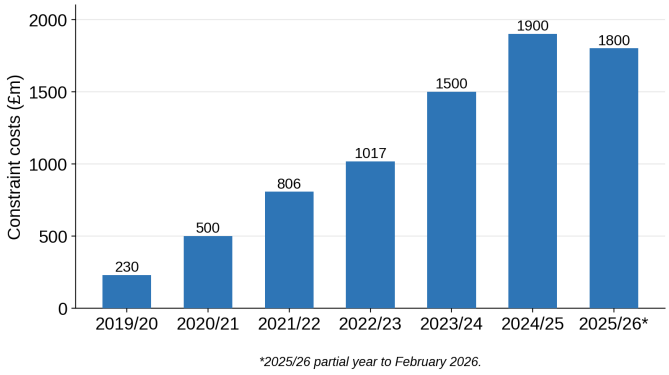


Figure 2. Annual transmission constraint costs in Great Britain, 2019/20–2025/26, adapted from [42]

4.4 Ageing Infrastructure and Delivery Lead Times

Much of the existing transmission fleet dates from the 1960s and 1970s and is nearing the end of its design life [39, 44]. Under the pre-reform framework, average connection lead times for large new generation projects reached 10–15 years, and the Ofgem end-to-end review reported project cost inflation of up to 200 percent for schemes affected by connection delays [34]. Persistent delivery risk reflecting supply chain constraints, planning bottlenecks and skills shortages has been a further driver of institutional reform [21, 45].

5 Architecture of Reform

The GB reform programme can be organised around three mutually reinforcing pillars: (i) redesign of the connections process; (ii) acceleration of transmission investment; and (iii) strengthening of the regulatory accountability regime. Table 1 summarises the principal instruments in each pillar.

Table 1. Principal reform instruments in Great Britain, 2022–2026

Pillar	Instrument	Owner	Purpose
Connections	Target Model Option 4+ (TMO4+) (Gate 1/Gate 2)	National Energy System Operator (NESO) / Office of Gas and Electricity Markets (Ofgem)	Replace "first come, first served" with "first ready and needed"
Connections	Gate 2 to Whole Queue	NESO	Reorder existing queue against readiness and strategic alignment
Investment	Accelerated Strategic Transmission Investment (ASTI) framework	Ofgem	Fast-track funding for 17 major transmission projects
Investment	Advanced Procurement Mechanism	Ofgem	Enable early procurement of long-lead items (high-voltage direct current (HVDC) cables, transformers)
Investment	RIIO-T3 price control (2026–2031)	Ofgem	Provide up to £35bn baseline + pipeline funding envelope
Regulation	End-to-end connections review	Ofgem	Introduce enforceable delivery obligations and penalties
Regulation	Strategic Spatial Energy Plan	NESO/Department for Energy Security and Net Zero (DESNZ)	Whole-system spatial planning aligned with 2030–2050 pathways

Note: Compiled by the author from [10, 11, 28, 30–35].

5.1 Reform of the Connections Process

The most visible reform concerns the connections process itself. Under the previous "first come, first served" regime, any developer could apply at any time and receive a place in the queue regardless of project maturity or system need. Speculative applications from projects with no realistic delivery prospect accumulated in the queue, blocking timely access for shovel-ready projects [11]. The Target Model Option 4+ (TMO4+), which went live on 10 June 2025 following Ofgem approval in April 2025, introduces gated application windows and requires applicants to satisfy readiness and strategic-alignment criteria before receiving a firm connection offer [46].

Figure 3 provides a simplified representation of the TMO4+ process. Projects that cannot yet satisfy readiness criteria (for example, land rights or planning consent) may be issued a Gate 1 indicative offer, but do not hold a position in the delivery pipeline until they meet Gate 2. Projects satisfying both readiness and strategic-alignment criteria proceed to Gate 2 and receive a firm offer with a place in the reordered delivery pipeline confirmed by NESO in December 2025 [35, 46].

An early quantitative outcome, reported as of December 2025, is that approximately three thousand applications have been re-assessed through the process, unlocking 283 GW of generation and storage and 99 GW of transmission-connected demand for the reordered pipeline, while removing speculative applications with no realistic prospect of delivery [35]. Separately, and outside the three reform pillars examined here, the Connections Accelerator Service (CAS) constitutes a mechanism within the current regulatory framework, introduced by Government as announced in the Industrial Strategy (June 2025) to support projects of strategic importance such as AI Growth Zones (AIGZ) [10, 47, 48]

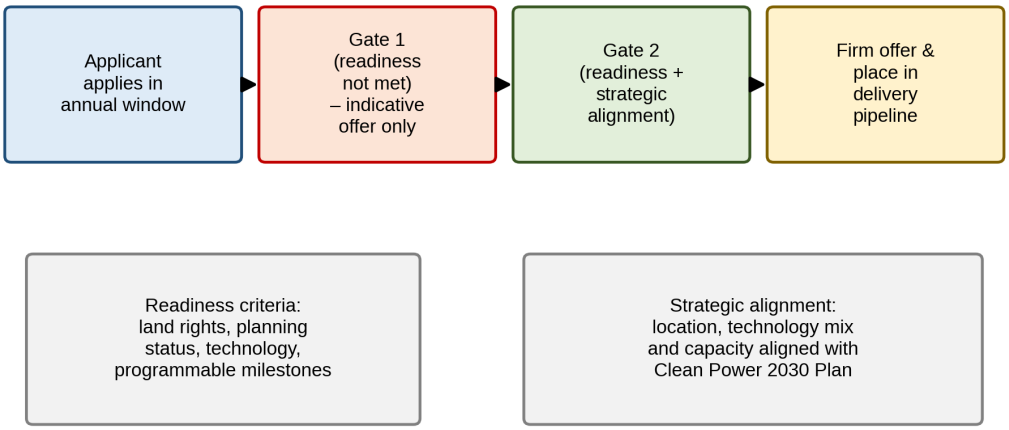


Figure 3. Simplified representation of the Target Model Option 4+ (TMO4+) gated connections process, adapted from [35, 46]

5.2 Acceleration of Transmission Investment

The second pillar is a step change in transmission investment. The ASTI framework, introduced by Ofgem in late 2022, identified 26 network projects considered critical to decarbonisation and streamlined their funding, procurement and consenting arrangements [33]. Under ASTI, National Grid Electricity Transmission (NGET) has organised 17 major projects including four Eastern Green Link HVDC subsea cables, Sea Link, and a series of overhead-line reinforcements under a single "Great Grid Upgrade" programme [38, 49].

The RIIO-T3 price control, running from April 2026 to March 2031, provides the funding framework. In its final determination published in December 2025, Ofgem approved a baseline expenditure of £10.3 billion, with additional pipeline funding released progressively as projects mature [32]. NGET's business plan envisages up to £35 billion of investment across the five-year period, including around £24 billion in pipeline projects and £15 billion specifically to expand network capacity [38]. Figure 4 illustrates the indicative allocation.

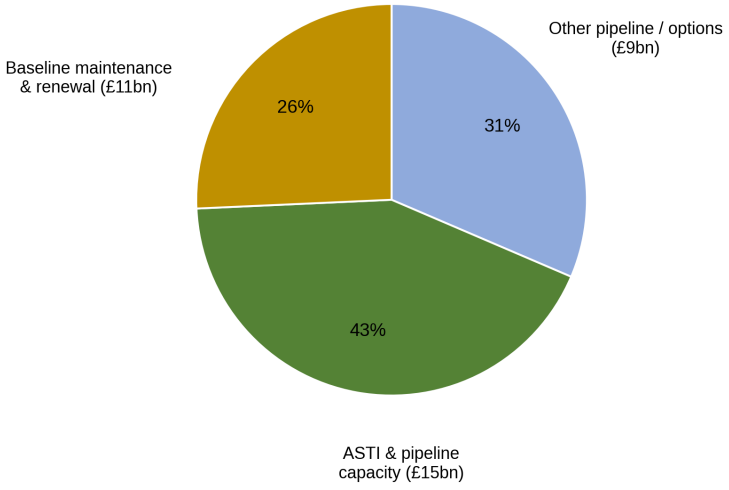


Figure 4. Indicative allocation of National Grid Electricity Transmission (NGET) RIIO-T3 investment envelope, adapted from [38]

Note: ASTI = Accelerated Strategic Transmission Investment.

Beyond RIIO-T3, in its 2026 Beyond 2030 study NESO projected further investment requirements of up to £89 billion in the period beyond 2030, an increase of approximately 50 percent on the 2024 estimate, driven by inflation and the accelerated build-out required by Clean Power 2030 [36]. Table 2 places the GB commitment in comparative context.

Table 2. Comparative scale of near-term electricity network investment plans

Jurisdiction	Programme	Period	Indicative Envelope	Reference
Great Britain	RIIO-T3 National Grid Electricity Transmission (NGET)	2026–2031	Up to £35 bn	[38]
Great Britain	Beyond 2030 pipeline	2030–2035	Up to £89 bn	[36]
European Union	European Court of Auditors (ECA) estimate to 2050	2025–2050	∼€2 tn	[21]
United States	Department of Energy (DOE) grid deployment	2022–2032	US$65 bn (Infrastructure Investment and Jobs Act, IIJA)	[24]
Global	International Energy Agency (IEA) reference case	2023–2030	US$600 bn/year	[3]

Note: Envelopes are not strictly comparable owing to differing scope, timeframes and price bases; they are reported to indicate order of magnitude.

5.3 Strengthening the Regulatory Accountability Regime

The third pillar is the regulatory framework. Ofgem's end-to-end review, updated in December 2025, identified an "asymmetry of risk" between developers and network companies: developers bore very substantial sunk costs from missed connection dates while network operators faced minimal financial consequences [34]. The review therefore proposed tougher licence obligations, standardised connection processes, financial penalties for missed milestones, and enhanced enforcement powers [34, 50]. Consultation on the demand connections regime opened in November 2025 [51].

These reforms are complemented by whole-system planning instruments. The Strategic Spatial Energy Plan, developed by NESO with the United Kingdom, Scottish and Welsh governments, seeks to co-optimise generation, storage and network build across the country [35, 52]. In parallel, the Planning and Infrastructure Act creates statutory powers for the designation and prioritisation of nationally significant energy projects [53].

Figure 5 synthesises the analysis above by depicting the three reform pillars not as parallel workstreams but as an interdependent system. Connections reform (TMO4+, Gate 2 to Whole Queue) determines which projects are ready to draw on new network capacity; transmission investment (ASTI, the Advanced Procurement Mechanism, RIIO-T3) determines how quickly that capacity becomes available; and the regulatory accountability regime (the end-to-end connections review, the Strategic Spatial Energy Plan) supplies the enforcement and spatial-coordination mechanisms that keep the other two pillars aligned with the Clean Power 2030 target. The framework illustrates that a shortfall in any single pillar propagates to the other two: for example, a connections queue reordered under TMO4+ without matching transmission capacity under RIIO-T3 would simply relocate the bottleneck from the queue to the network itself.

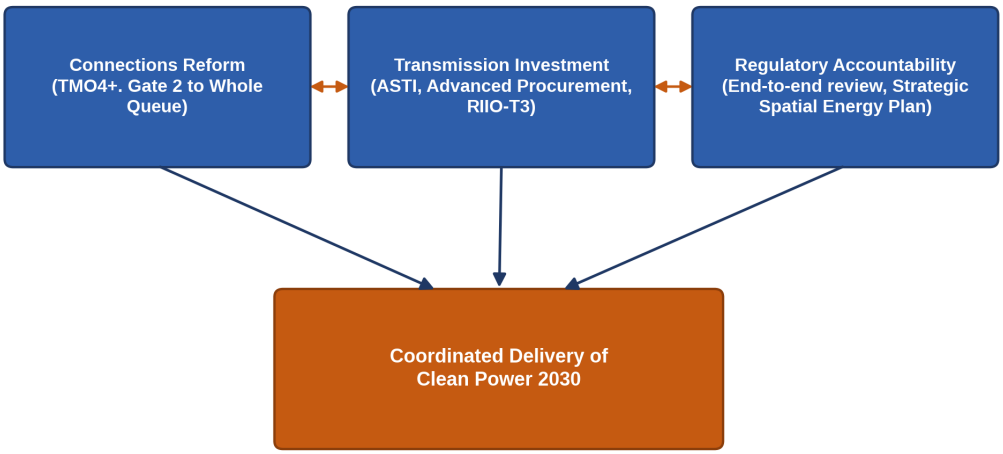


Figure 5. Conceptual framework linking the three Great Britain grid reform pillars
Note: TMO4+ = Target Model Option 4+; ASTI = Accelerated Strategic Transmission Investment.

6 Cross-Cutting Lessons and Comparative Perspective

Three cross-cutting lessons emerge from the analysis. First, the reforms confirm that grid transformation is as much an institutional as a technical challenge. The technical solutions HVDC subsea cables, dynamic line ratings, and offshore hybrid assets have been available for more than a decade [14, 54]. What has changed is the emergence of a coordinated architecture in which the system operator, the regulator, government and network companies each hold defined roles, and in which strategic planning is explicitly aligned with policy targets [35, 51].

Second, the interdependence of the three reform pillars is essential. Connections reform without adequate investment would simply move the queue elsewhere. Investment without connections reform would fund infrastructure for projects that may never materialise. Regulation without either would preserve the status quo. The GB experience underscores the importance of designing reforms as a package [55].

Third, cost allocation and consumer protection remain unresolved. As set out in its December 2025 RIIO-T3 final determination, Ofgem estimates that the RIIO-T3 investment programme will unlock up to £90 billion of transmission investment and could contribute to avoided constraint costs of around £12 billion over the price control period [32, 38]. Yet consumers will bear the up-front financing costs through network charges, and reforms to the regulatory allowance framework will therefore need to be paired with continued vigilance on efficiency and value for money [21, 56].

The European Union confronts many of the same challenges. The ECA has concluded that the current pace of planned grid investment lags the estimated requirements of climate neutrality, and that permitting, equipment supply chains and skilled labour are common bottlenecks [21]. Emerging solutions including interoperable offshore grids, flexibility markets, digital twins and cross-border coordination through European Network of Transmission System Operators for Electricity (ENTSO-E) closely mirror those pursued in GB [18, 57]. The convergence suggests that many of the GB reforms are portable, though the specific institutional design must be adapted to each jurisdiction's legal and market context.

These generalisations nonetheless require qualification. GB benefits from a single, technically neutral system operator—NESO, responsible for whole-system planning, and from a mature, independent economic regulator (Ofgem) empowered to set enforceable licence conditions across a small number of unbundled transmission owners [30–32]. These institutional preconditions do not hold universally. In vertically integrated or state-owned utility markets, and in federal systems where transmission planning and cost allocation are divided across multiple regulatory authorities, such as the interaction between the Federal Energy Regulatory Commission and individual state public utility commissions in the United States, an equivalent single gated connections process or unified price-control settlement may not be institutionally or legally feasible without prior market restructuring [58]. Similarly, in emerging and developing economies where connection queues are driven less by an oversupply of speculative applications and more by constrained access to capital, skilled labour or imported grid equipment, the binding constraint on reform differs materially from the GB case, and reform sequencing would need to prioritise financing and capacity-building measures alongside, or ahead of, procedural reform of the connections process. The GB experience is therefore best read as a demonstration of a coordinated reform architecture rather than a template whose specific instruments can be transplanted unmodified.

7 Policy Implications

Four policy implications follow from the analysis. First, decision-makers should treat connections reform, transmission investment and regulatory accountability as a single design problem rather than sequential initiatives. In the GB case, delays in any one pillar would have undermined the others. Second, a credible strategic plan anchored in a legally significant policy target here, Clean Power 2030 provides the reference point against which project readiness and network need can be assessed [10, 35]. Third, delivery risk must be actively managed through mechanisms such as early construction funding, advanced procurement and enforceable milestones [33]. Fourth, transparency about what reform does and does not deliver is essential: sponsorship and prioritisation instruments must clearly communicate that inclusion in a strategic list does not guarantee acceleration [47].

8 Conclusions

GB has embarked on the most substantial reform of its electricity network in a generation. The evidence assembled here shows that reforms initiated between 2022 and 2026 have begun to unblock a connections queue that had grown to more than 700 GW of generation and storage, have committed public and consumer resources to a step change in transmission investment on the scale of £35 billion under RIIO-T3, and have redesigned the regulatory accountability framework governing network delivery. Full evaluation will only be possible once the new pipeline is delivered through the late 2020s and early 2030s. Nonetheless, the case supports three broader propositions. Grid reform is inseparable from decarbonisation; it must be designed as a coordinated package spanning process, investment and regulation; and its success depends on institutional coherence as much as on technical innovation.

Future research should track the delivered lead-time reductions, constraint cost trajectories, and consumer bill impacts of the reformed system. In particular, establishing a longitudinal monitoring framework, for example an

annual panel tracking connection lead times, constraint costs and delivered network capacity against the milestones set out in the RIIO-T3 price control and the Strategic Spatial Energy Plan, would allow researchers and regulators to distinguish transitional implementation effects from durable structural improvement over the 2026–2031 price-control period and beyond. Such monitoring could be strengthened by drawing on operational grid data, including wide-area monitoring and synchrophasor measurements increasingly deployed across the transmission network, to provide a real-time, quantitative evidence base for evaluating whether reformed connections and investment processes are translating into measurable improvements in network utilisation and constraint management, rather than relying solely on the periodic, retrospective indicators used in this study. Quantitative modelling of the carbon-intensity and lifecycle sustainability impacts of the accelerated transmission build programme would strengthen the evidence base. Comparative analysis extending to jurisdictions pursuing similar reforms in continental Europe, North America, and Asia would help distinguish context-specific from universally applicable lessons.

Data Availability

The data supporting the analysis in this article are drawn entirely from publicly available policy documents, regulatory determinations, and network operator publications cited in the References.

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Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI tools were used to assist in formatting the manuscript to the journal template. All intellectual content, analysis, and conclusions are the sole responsibility of the author. AI-generated content was reviewed and verified by the author prior to submission.

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